



**Guangdong Technion**

Israel Institute of Technology

广东以色列理工学院



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# **Tutorial Course for Intro to Sensor-Integrated Systems**

Qilong Zhong, MSc

Mechanical Engineering and Robotics

Guangdong Technion - Israel Institute of Technology

qilong.zhong@gtiit.edu.cn

2026

# Digital-to-Analog Conversion (DAC)

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## 1. Summing Amplifier DAC (Weighted Resistor Network)

- **Principle:** Utilizing an operational amplifier to sum the currents corresponding to different bits according to their binary weights.
- **Output Voltage Formula (Example for an 8-bit DAC):**

$$V_{OUT} = -\frac{R_F}{R} V_{REF} \left[ \frac{b_7}{2^0} + \frac{b_6}{2^1} + \dots + \frac{b_0}{2^7} \right]$$

Where  $b_i$  is the binary bit (1 or 0),  $R_F$  is the feedback resistor, and  $V_{REF}$  is the reference voltage.

## 2. Relationship between Resolution and Output Voltage

- **Resolution:** The voltage step change caused by a change in the Least Significant Bit (LSB).

$$Resolution = \frac{V_{OUT,MAX}}{2^n - 1}$$

- **Output Calculation Shortcut:**

$$V_{OUT} = Resolution \times (Decimal\ value\ of\ the\ input\ signal)$$

## Exercise

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The digital-to-analogue converter of Figure 10.6(a) is required to give an output voltage in the range 0 to 2.55 V, corresponding to an 8-bit digital input signal 00000000 to 11111111.

- (a) Assuming  $V_{\text{REF}} = -15 \text{ V}$ ,  $R = 1 \text{ k}\Omega$ , calculate the value of  $R_F$  required.
- (b) Find the output voltage corresponding to an input signal of 11000101.

# Solution

(a) Given  $V_{REF} = -15V$ ,  $R = 1\text{ k}\Omega$ , find  $R_F$ :

- When the input is all 1s, the maximum output  $V_{OUT,MAX} = 2.55V$  is obtained.

Substituting into the formula:

$$2.55 = -\frac{R_F}{1000}(-15) \left( 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{128} \right)$$

- The sum of the geometric series inside the parentheses is  $\frac{255}{128}$ :

$$2.55 = \frac{15 \cdot R_F}{1000} \times \frac{255}{128}$$

- Solving for  $R_F$ :  $R_F = \frac{2.55 \times 128000}{15 \times 255} = \frac{256}{3} \approx \mathbf{85.33\ \Omega}$

(b) Find the output voltage for input **11000101** :

- Calculate Resolution:  $\frac{2.55\text{ V}}{2^8 - 1} = \frac{2.55}{255} = 0.01\text{ V/step}$
- Convert **11000101** to decimal:  $128 + 64 + 4 + 1 = 197$
- Output Voltage:  $V_{OUT} = 0.01\text{ V} \times 197 = \mathbf{1.97\text{ V}}$

# Frequency-to-Digital Conversion

## 1. Measurement Principle

➤ Utilizes two sets of counters working in tandem:

- **Signal Counter:** Records the number of complete periods ( $N_s$ ) of the low-frequency signal to be measured.
- **Clock Counter:** Records the number of pulses ( $N_c$ ) of the high-frequency reference clock within that total time interval.

## 2. Core Formulas

➤ The total measurement time  $t$  is equivalent:

$$t = N_s \times T_{sig} = N_c \times T_{clk}$$

$$N_c = N_s \times T_{sig} \times f_{clk} = N_s \times \frac{f_{clk}}{f_{sig}}$$

## 3. Overflow Prevention Design Principles (Worst-case scenario)

➤ Counters have maximum capacity limits (e.g.,  $2^{16} - 1 = 65535$  for a 16-bit counter).

➤ **When the worst case occurs:** When the frequency of the signal being measured  $f_{sig}$  is at its lowest, the total measurement time is at its longest, making the clock counter most prone to overflow. The limits must be calculated based on this scenario.

## Exercise

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The output signal of a vortex flowmeter has a frequency range between 5 and 50 Hz. The mean period of the signal is to be measured using a signal counter and a clock counter. The signal is input to the signal counter, which increments until  $N_s$  cycles have been counted. The clock counter is 16-bit binary and has a clock input of frequency 10 kHz. The clock counter increments until a count of  $N_s$  is registered on the signal counter. What is the maximum possible value of  $N_s$ ?

# Solution

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## 1. Define Limit Conditions

- The maximum count value for a 16-bit counter is  $N_{c(max)} = 2^{16} - 1 = 65535$ .
- The condition most likely to cause an overflow is when the total measurement time is longest, which occurs at the lowest signal frequency ( $f_{sig(min)} = 5 \text{ Hz}$ ).
- In this case, the maximum signal period  $T_{sig(max)} = 1/5 = 0.2 \text{ s}$ .

## 2. Construct Inequality

- To ensure no overflow occurs, the number of clock pulses must satisfy:

$$N_c = N_s \times T_{sig(max)} \times f_{clk} \leq 65535$$

## 3. Solve for $N_s$ :

$$N_s \times 0.2 \text{ s} \times 10000 \text{ Hz} \leq 65535$$

$$N_s \times 2000 \leq 65535$$

$$N_s \leq 32.7675$$

## 4. Conclusion:

- Since  $N_s$  represents the number of complete signal periods to be counted, it must be an integer. Rounding down, the maximum value is  $N_s = 32$ .

## Homework 1\_1

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A platinum resistance sensor is used to interpolate between the triple point of water (0 °C), the boiling point of water (100 °C) and the freezing point of zinc (419.6 °C). The corresponding resistance values are 100.0  $\Omega$ , 138.5  $\Omega$  and 253.7  $\Omega$ . The algebraic form of the interpolation equation is:

$$R_T = R_0(1 + \alpha T + \beta T^2)$$

where  $R_T \Omega$  = resistance at  $T$  °C  
 $R_0 \Omega$  = resistance at 0 °C  
 $\alpha, \beta$  = constants.

Find the numerical form of the interpolation equation.



# Solution

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## Solution

Point 1 (Triple point of water):  $T_1 = 0, R_{T1} = 100$

Point 2 (Boiling point of water):  $T_2 = 100, R_{T2} = 138.5$

Point 3 (Freezing point of Zinc):  $T_3 = 419.6, R_{T3} = 253.7$

## Solve for Constant $R_0$

Substitute the data from **Point 1** into the general equation:

$$100 = R_0(1 + \alpha \cdot 0 + \beta \cdot 0^2)$$

It can be directly concluded that:

$$R_0 = 100\Omega$$

## Substitute Point 2 (100 °C, 138.5 $\Omega$ ):

$$138.5 = 100(1 + \alpha \cdot 100 + \beta \cdot 100^2)$$

$$\alpha \cdot 100 + \beta \cdot 100^2 = 0.385$$

## Substitute Point 3 (419.6 °C, 253.7 $\Omega$ ):

$$253.7 = 100(1 + \alpha \cdot 419.6 + \beta \cdot 419.6^2)$$

$$\alpha \cdot 419.6 + \beta \cdot 419.6^2 = 1.537$$

$$\beta = -5.8507 \times 10^{-7}, \alpha = 3.9085 \times 10^{-3}$$

$$R_T = 100(1 + 3.9085 \times 10^{-3}T - 5.8507 \times 10^{-7}T^2)$$

## Homework 1\_2

The following results were obtained when a pressure transducer was tested in a laboratory under the following conditions:

- I Ambient temperature 20 °C, supply voltage 10 V (standard)
- II Ambient temperature 20 °C, supply voltage 12 V
- III Ambient temperature 25 °C, supply voltage 10 V

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|              |   |     |      |      |      |    |
|--------------|---|-----|------|------|------|----|
| Input (barg) | 0 | 2   | 4    | 6    | 8    | 10 |
| Output (mA)  |   |     |      |      |      |    |
| I            | 4 | 7.2 | 10.4 | 13.6 | 16.8 | 20 |
| II           | 4 | 8.4 | 12.8 | 17.2 | 21.6 | 28 |
| III          | 6 | 9.2 | 12.4 | 15.6 | 18.8 | 22 |

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- (a) Determine the values of  $K_M$ ,  $K_I$ ,  $a$  and  $K$  associated with the generalised model equation  $O = (K + K_M I_M)I + a + K_I I_I$ .
- (b) Predict an output value when the input is 5 barg,  $V_S = 12$  V and ambient temperature is 25 °C.

## Homework 2\_1

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1. A temperature sensor sensing element have ideal transfer function  $R = R_0(1 + T)$ , and then the signal condition element have ideal transfer function as  $V_{Th} = V_{Th0}(1 + 1.08R)$ . With  $R_0 = 100 \Omega$  and  $V_{Th0} = 1.2 V$ . But due to nonlinearity, they have transfer function as  $R = R_0(1 + T - 0.2T(T - 1))$  and  $V_{Th} = V_{Th0}(1 + 1.08R + 0.003(R - R_0)(R - 2R_0))$ . Please plot the output  $V_{Th}$  as a function of input  $T$ , considering the following four combinations with input  $T \in [0.01, 1]$  K:

- A) Ideal sensing element + ideal singal condition element;
- B) Nonlinear sensing element + ideal singal condition element;
- C) Ideal sensing element + nonlinear singal condition element;
- D) Nonlinear sensing element + nonlinear singal condition element.

Please explain why case D) can have better performance than B) and C).

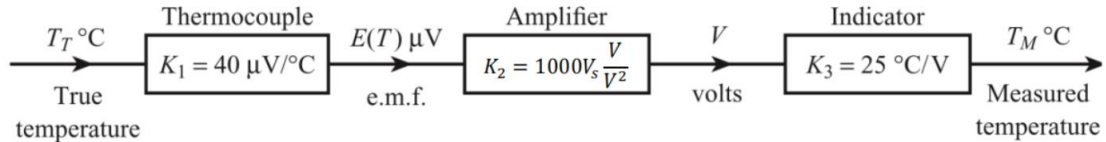
## Homework 2\_2

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2. A sensor has transfer function as  $O = (K + K_M I_M)I + a + N(I) + K_I I_I = (2 + 0.4 I_M)I + 4 + 0.4 I_I$ . Please design compensation element so that after the compensation element in series, the environment influences are gone.

## Homework 2\_3

3. The sensor shown below have  $K_2 = 1000V_s \frac{V}{V^2}$ , i.e., the amplification coefficient is influenced by the voltage supply  $V_s$ . Initially,  $V_s$  is equal to  $1 V$ .  $T_T = 20^\circ\text{C}$



A) Please calculate the system error when the  $V_s = 1.2 V$ .

B) Please add high gain feedback with coefficient of 50 and then add another block with coefficient of 50 after  $T_M$  to obtain the final output  $T'_M$ , calculate the system error again when the  $V_s = 1.0 V$  and  $V_s = 1.2 V$ .

# Homework 2\_4

A temperature measurement system consists of a thermocouple, an amplifier, an 8-bit analogue-to-digital converter and a microcontroller with display facilities. Table Prob. 5 gives the model equations and parameters for each element in the system. The temperature of the thermocouple measurement junction is  $T_1$  °C and the temperature of the reference junction is  $T_2$  °C. The microcontroller corrects for  $T_2$  having a non-zero mean value.

- Estimate the mean and standard deviation of the error probability density function when the input temperature  $T_1$  is 100 °C. Treat rectangular distribution as normal with  $\sigma = h/\sqrt{3}$ .
- Explain briefly what modifications should be made to the system to reduce the quantities calculated in (a).

|                           | Thermocouple                                                                                                       | Amplifier                                                   | Analogue-to-digital converter                                                          | Microcontroller with display                 |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------|
| Model equations           | $E_{T_1, T_2} = a_0 + a_1(T_1 - T_2) + a_2(T_1^2 - T_2^2)$                                                         | $V = K_1 E_{T_1, T_2} + b_1$                                | $n = K_2 V + b_2$                                                                      | $T_m = K_3 n + b_3$                          |
| Mean values               | $\bar{a}_0 = 0$<br>$\bar{a}_1 = 4.3796 \times 10^{-2}$<br>$\bar{a}_2 = -1.7963 \times 10^{-5}$<br>$\bar{T}_2 = 20$ | $\bar{K}_1 = 255$<br>$\bar{b}_1 = 0.0$                      | $\bar{K}_2 = 0.1$<br>$\bar{b}_2 = 0.0$<br>$n$ rounded to nearest integer               | $\bar{K}_3 = 1.0$<br>$\bar{b}_3 = 20$        |
| Statistical distributions | Normal with<br>$\sigma_{a_0} = 0.05$<br>$\sigma_{T_2} = 2.0$<br>$\sigma_{a_1} = \sigma_{a_2} = 0.0$                | Normal with<br>$\sigma_{b_1} = 5.0$<br>$\sigma_{K_1} = 0.0$ | $b_2$ has a <b>rectangular distribution</b> of width $\pm 0.5$<br>$\sigma_{K_2} = 0.0$ | $\sigma_{K_3} = 0.0$<br>$\sigma_{b_3} = 0.0$ |